

PAPER_1550 — Compton Wavelength $\lambda_C = \lambda_C = K_{MEX} + F_{TRZ} \cdot D_{phys} - F_{TRZ} \cdot SSQ = 2.083 + 0.4 - 0.057 = 2.426$ (lead)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Quantum/Atomic **Date:** June 18, 2026 **Status:** CLOSED — 0.015% closure

Observation

PAPER_1209DD_S620 (Quantum/Atomic Unified Proof Set) lists **Compton Wavelength λ_C** as a closure with the formula:

Compton Wavelength $\lambda_C = \lambda_C = K_{MEX} + F_{TRZ} \cdot D_{phys} - F_{TRZ} \cdot SSQ = 2.083 + 0.4 - 0.057 = 2.426$ (lead)

Status tier: **0.015%**. Units: lead.

UQFF Closed Identity

$\lambda_C = K_{MEX} + F_{TRZ} \cdot D_{phys} - F_{TRZ} \cdot SSQ = 2.083 + 0.4 - 0.057 = 2.426$ (lead) 0.015%

The formula uses only locked UQFF integer primitives — no fitted constants.

Physical Interpretation

Compton Wavelength λ_C is a fundamental quantum/atomic constant. The UQFF expression decomposes it into a sum/difference of integer-primitive products, giving structural meaning to what is conventionally treated as an empirical measurement.

NOT REPLACEMENT

CODATA/PDG treats this constant as a precision measurement. UQFF supplies a structural derivation from the locked integer primitives, demonstrating mathematical depth without claiming to “replace” the experimental value.

Reference

- Source: PAPER_1209DD_S620
- Related: PAPER_1216 (45-constant cascade master index)
- Calculator dispatch: `calculate_paradox({"paradox": "compton_lambda_2_426"})`

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